

Multiple Choice (10 marks)

1. Neural networks:
 - (a) Optimize a convex objective function
 - (b) Can only be trained with stochastic gradient descent
 - (c) Can use a mix of different activation functions
 - (d) None of the above
2. __ refers to a model that can neither model the training data nor generalize to new data.
 - (a) good fitting
 - (b) overfitting
 - (c) underfitting
 - (d) all of the above
3. Which of the following sentence is FALSE regarding regression?
 - (a) It relates inputs to outputs.
 - (b) It is used for prediction.
 - (c) It may be used for interpretation.
 - (d) It discovers causal relationships
4. What would you do in PCA to get the same projection as SVD?
 - (a) Transform data to zero mean
 - (b) Transform data to zero median
 - (c) Not possible
 - (d) None of these
5. A and B are two events. If $P(A, B)$ decreases while $P(A)$ increases, which of the following is true?
 - (a) $P(A|B)$ decreases
 - (b) $P(B|A)$ decreases
 - (c) $P(B)$ decreases
 - (d) All of above

True / False (10 marks)

Answer True or False

6. True or False: Caltech-101 includes a larger number of images than ImageNet, providing a richer resource for learning algorithms.
7. True or False: Random crop and horizontal flip are common image data augmentation techniques used in natural images.
8. True or False: The number of test examples required does not depend on the error rate, and remains constant regardless of its value.
9. True or False: DenseNets typically require more memory than ResNets due to their dense connectivity pattern.
10. True or False: Ridge regression is preferred for feature selection because it reduces model complexity by eliminating variables.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Explain the joint probability of H, U, P, and W in the context of the given Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$. Discuss the roles of conditional probabilities in this network.
12. Discuss the existential risks associated with artificial intelligence as articulated by Stuart Russell, analyzing their implications for the future of AI development and safety.

End of Paper